PS 2

$$E = \iint \frac{1}{2} B \kappa^2 \, dA + \iiint P \, dV$$

$$= \frac{1}{2} B \left(\frac{2}{R}\right)^2 \iint dA + P \iiint dV$$

$$= \frac{1}{2} B \left(\frac{2}{R}\right)^2 \frac{1}{2} 4\pi R^2 + \dots$$

AREA

HALF SPHERE

$$= \frac{8\pi B}{2} \frac{1}{R^2} R^2$$

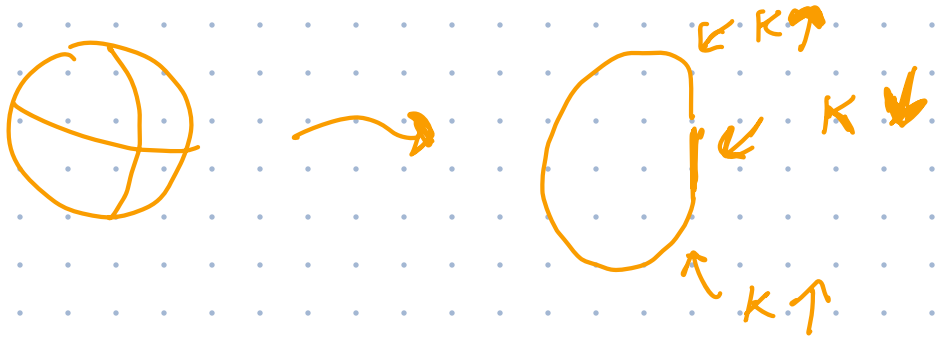


BENDING ENERGY

$$E = 8\pi B$$

TOTAL INTEGRATED CURVATURE IS INDEPENDENT OF RADIUS

GAUSS-BONNET THEOREM

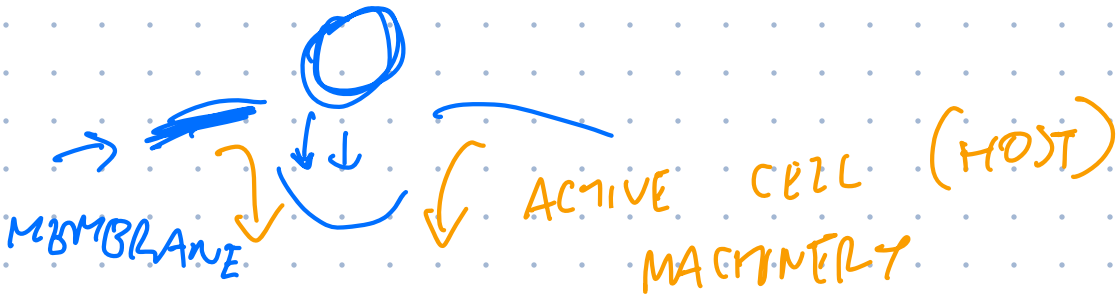


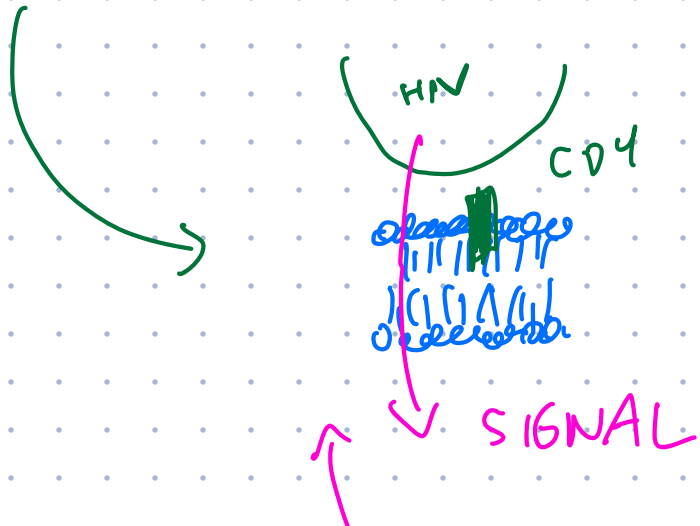
HOW DOES E COMPARE TO:

E_{ATP} ?. ← $E \gg 1\ E_{ATP}$

$k_B T$?. $E \sim 10^3\ E_{ATP}$ ✓✓✓✓

ii)





ACTIVE CELL MACHINERY

PART II : DIFFUSION